

Embarrassingly parallel workflows

06.03.2026

Embarrassingly Parallel workflows and how to execute them

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Embarrassingly parallel workflows

What?

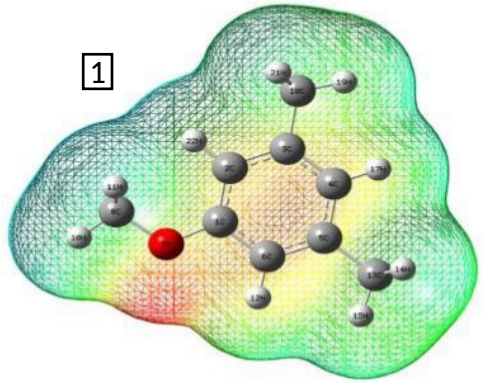
- Executing many independent small jobs simultaneously to use the hardware resource efficiently.

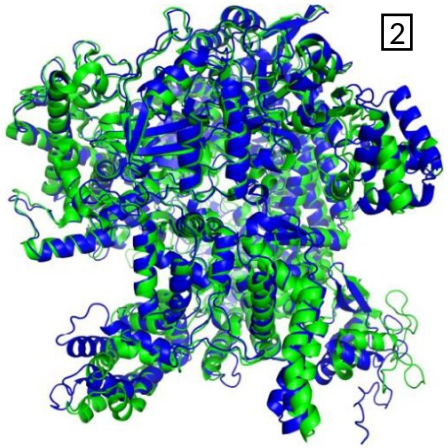
When?

- It is not possible to combine multiple runs/tasks into a single application.
- Large parameter sweep based on the same application or workflow with dependent pipeline steps.
- Workflow involves runs that depend on independent software which does not scale beyond a certain number of cores.

Why?

- Nodes are getting denser with respect to resources such as number of cores/threads, GPU cards, Memory.
- Applications do not necessarily scale even within a node: shared memory (OpenMP, MPI SHM, pthreads).





1. Mathiyalagan, A., Manimaran, K., Muthu, K., Rajakantham, M., Density functional theory study on the electronic structures and spectral properties of 3,5-Dimethylanisole dye sensitizer for solar cell applications, Results in Chemistry, Volume 3, 2021

2. Jumper, J., Evans, R., Pritzel, A. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583-589 (2021)

How?

Tools

- Linux background processes (within a node) along with **numactl**.
  - Supported on Snellius and requires prior knowledge of bash scripting and process placement using **numactl**.
  - Provides maximum manual control if one knows what to do.
- SLURM arrays (to perform similar jobs on multiple allocations).
  - Supported on Snellius and requires prior knowledge of SLURM array options.
  - Has some limitations.
- Services for execution of many jobs inside a single allocation, e.g. QCG-PilotJob (https://qcg-pilotjob.readthedocs.io/en/develop/).
  - Supported on Snellius and requires prior knowledge of Python scripting.





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How?

Tools

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There are many more tools















SLURM arrays

Limitations

- SLURM job arrays offer a mechanism for submitting and managing **collections of similar jobs**.
- All jobs must have the **same initial options** (e.g. size, time limit, etc.).
- The **max array size** depends on the settings defined in the SLURM configuration file. On Snellius it's **30000**.
- The **allocation size** per instance **cannot be below the minimum allocation** within SLURM. For instance, on Snellius it's 16 cores and 28 GiB of memory on the "rome" partition.
- One must **wait for each task** in the array to get allocated and finish.

job.sh

```
#!/bin/bash
#SBATCH -N 1
#SBATCH -n 2
#SBATCH -p rome
#SBATCH -t 1
hostname
```

```
$ sbatch --array=0-10 job.sh
```



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hostname
```

```
$ sbatch --array=0-10 job.sh
```

Will create 11 tasks!

Ways of control

- Jobs which are part of a job array will have the environment variable **SLURM_ARRAY_TASK_ID** set to its array index value.
- A **maximum number of simultaneously running tasks** from the job array may be specified **using a "%" separator**.

```
$ cat job.sh
#!/bin/bash
#SBATCH -N 1
#SBATCH -n 2
#SBATCH -p rome
#SBATCH -t 1
echo "SLURM_ARRAY_TASK_ID: $SLURM_ARRAY_TASK_ID"
hostname

$ sbatch --array=0-10 job.sh
...
$ sbatch --array=0-10%2 job.sh
...
```



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hostname

$ sbatch --array=0-10 job.sh
...
$ sbatch --array=0-10%2 job.sh
...
```

Returns a task ID

Will tell SLURM to submit at most 2 tasks at the same time

Ways of control

Useful environment variables

- **SLURM_ARRAY_JOB_ID** is the first job ID of the array.
- Each task finds its array index in **SLURM_ARRAY_TASK_ID**.
- **SLURM_ARRAY_TASK_COUNT** is the number of tasks in the job array.
- The highest index is in **SLURM_ARRAY_TASK_MAX**.
- **SLURM_ARRAY_TASK_MIN** gives the lowest index.



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Ways of control

File names

Two additional options are available to specify a job's **stdin**, **stdout**, and **stderr** file names:

- **%A** will be replaced by the value of **SLURM_ARRAY_JOB_ID**
- **%a** becomes the task index, **SLURM_ARRAY_TASK_ID**

```
#!/bin/bash
...
#SBATCH --output slurm-%A_%a.out
...
```



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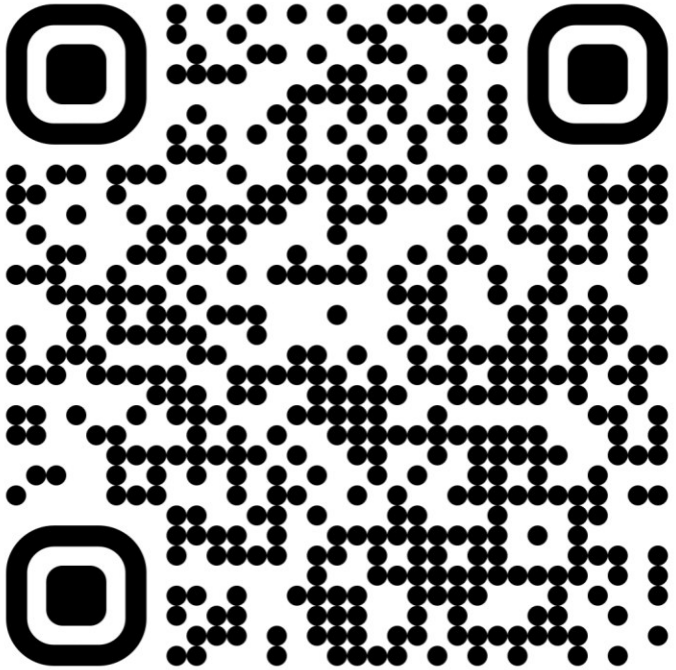
Hands-on

Hands-on

Clone the repository

- https://github.com/sara-nl/introduction-to-supercomputing

```
$ cd ~
$ git clone <repository_url.git>
$ cd introduction-to-supercomputing
$ cd course-module-qcg-pilotjob
```



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Hands-on #1

Task

- Submit a job array and print **hostname, TMPDIR, SLURM_ARRAY_JOB_ID**

Steps

- Go to **hands-on/example_01**
- Take a look at the job script **array_job.sh**
- Submit the SLURM job script, check the SLURM log file

Questions

- How many output files were created?
- Compare the host names.
- Note the names of **TMPDIR** that the jobs report.

```
#!/bin/bash
#SBATCH -N 1
#SBATCH -n 1
#SBATCH -p rome
#SBATCH -t 10
#SBATCH --array 0-11%3

echo "Running on "
hostname
echo "TMPDIR $TMPDIR"

echo "SLURM_ARRAY_JOB_ID $SLURM_ARRAY_JOB_ID"
echo "SLURM_ARRAY_TASK_ID $SLURM_ARRAY_TASK_ID"
echo "SLURM_ARRAY_TASK_COUNT $SLURM_ARRAY_TASK_COUNT"
echo "SLURM_ARRAY_TASK_MAX $SLURM_ARRAY_TASK_MAX"
echo "SLURM_ARRAY_TASK_MIN $SLURM_ARRAY_TASK_MIN"
```

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Hands-on #2

Task

- Compute the average of one column in 1000 CSV files
- Aggregate the results in one CSV file.

Steps

- Go to **hands-on/example_02**
- Take a look at the job script **array_job.sh**
- Run **prepare_input.sh**
- Submit the SLURM job script
- Run **aggregate.sh** when all jobs are finished

Questions

- What is the total execution time and SBU spent on this task?

```
#!/bin/bash
#SBATCH -N 1
#SBATCH -n 1
#SBATCH -p rome
#SBATCH -t 10
#SBATCH --array 0-999%10

module load 2024
module load Python/3.12.3-GCCcore-13.3.0

# copy input data and scripts to working directory
cp -r $SLURM_SUBMIT_DIR/input_mod $TMPDIR/
cp -r $SLURM_SUBMIT_DIR/average.py $TMPDIR/

cd $TMPDIR

echo "TMPDIR = $TMPDIR"

# average data
input_file_path=input_mod/${SLURM_ARRAY_TASK_ID}_*
echo "Analyzing file:  $input_file_path"
python3 average.py $input_file_path
```

QCG-PilotJob

QCGPilot-Job

Job manager inside SLURM

- Submits jobs within a single large allocation across multiple nodes.
- Does not have to wait for small individual allocations like in SLURM arrays.
- Optimizes resources for each job based on requirements such that the queue is always full.
- Possibility of restarting failed/timed out jobs.
- One can also specify complex result-based job dependencies.

Some caveats and drawbacks

- One needs to estimate, the total time of all the jobs combined.
- Job/system state files get produced in the submission directory.
- Might bring some complexity into execution of a Singularity containers with embedded MPI-based codes.
- When problems arise, the log files may say little.

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Workflow



Describes resources, loads the QCG-PilotJob module and runs the QCG-PilotJob python script

Invokes the QCG-PilotJob Manager that aggregates all your jobs along with resources and conditions based on which jobs need to be invoked

This is the main application that the jobs execute

Job output with QCG-PilotJob metrics

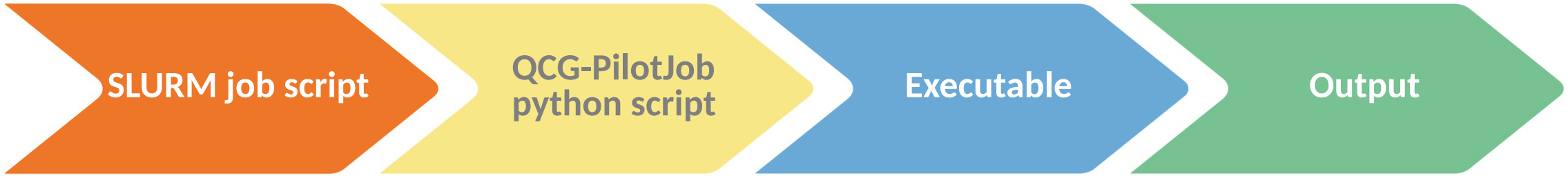
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QCGPilot-Job: Set up the environment

- The **QCG-PilotJob** package is available as a module in the **2024** environment.
- Note that the environment from the login node is ***not transferred*** to the compute nodes, unless defined in, e.g., the **.bashrc** file. Thus, you have to re-load the QCG-PilotJob package in your job script.

```
$ module purge
$ module load 2024
$ module load QCG-PilotJob/0.14.1-gfbf-2024a
```

Workflow



Describes resources, loads the QCG-PilotJob module and runs the QCG-PilotJob python script

Invokes the QCG-PilotJob Manager that aggregates all your jobs along with resources and conditions based on which jobs need to be invoked

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Job output with QCG-PilotJob metrics

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QCGPilot-Job: SLURM job script

run.sh

```
#!/bin/bash

#SBATCH -N 1
#SBATCH -n 16
#SBATCH -p rome
#SBATCH -t 10

module load 2024
module load QCG-PilotJob/0.14.1-gfbf-2024a

python test.py
```

Workflow



Describes resources, loads the QCG-PilotJob module and runs the QCG-PilotJob python script

Invokes the QCG-PilotJob Manager that aggregates all your jobs along with resources and conditions based on which jobs need to be invoked

This is the main application that the jobs execute

Job output with QCG-PilotJob metrics

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QCGPilot-Job: Python script

- Working with **QCG-PilotJob** takes a few mandatory steps:
  - Creating an instance of the **QCG-PilotJob Manager**
    - creation of a class object
  - Waiting for all submitted jobs to finish
    - will wait until some or all submitted to the **QCG-PilotJob Manager** jobs finish
  - Calling for the **finish()** method at the end of the Python script
    - this method terminates the background thread in which the instance of a **QCG-PilotJob Manager** has been run

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QCGPilot-Job: Python script

test.py

```python
from qcg.pilotjob.api.manager import LocalManager

manager = LocalManager()
print('available resources: ', manager.resources())
manager.wait4all()
manager.finish()
```

Example output

```
available resources: {'total_nodes': 1, 'total_cores': 96, 'used_cores': 0,
'free_cores': 96}
```

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QCGPilot-Job: Python script

Create an object

Wait for all jobs to finish

Finalize the execution

test.py

```
from qcg.pilotjob.api.manager import LocalManager

manager = LocalManager()
print('available resources: ', manager.resources())
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Example output

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available resources: {'total_nodes': 1, 'total_cores': 96, 'used_cores': 0,
'free_cores': 96}
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test.py

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```

Create an object

Wait for all jobs to finish

Finalize the execution

Example output

```
available resources: {'total_nodes': 1, 'total_cores': 96, 'used_cores': 0,
'free_cores': 96}
```

Available cores

Total number of nodes and cores within an allocation

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QCGPilot-Job: Create jobs

- To execute multiple jobs within a single SLURM allocation we need to:
  - Create an object of the **Jobs** class
  - Populate it with some parameters (describe jobs)
  - Submit the created object to the **QCG-PilotJob Manager**

```
...
from qcg.pilotjob.api.job import Jobs

jobs = Jobs()
jobs.add(...)
jobs_ids = manager.submit(jobs)
print('submited jobs: ', str(job_ids))
...
```

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QCGPilot-Job: Job description (main parameters)

- **name** - the job name
- **exec** - path to the executable program
- **args** - arguments for the program
- **script** - bash script content
- **stdin** - path to a file whose content goes to the standard input stream
- **stdout** - file that receives the standard output
- **stderr** - where the error stream is saved
- **modules** - list of modules to load before the job starts
- **numCores** - number of required cores
- **numNodes** - how many nodes the job needs
- **iteration** - iterations definition
- **model** - model of execution

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QCGPilot-Job: Job description (main parameters)

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QCGPilot-Job: execution models

- **default** - only a single process is launched within the allocation
- **threads** - is designed for running OpenMP tasks on a single node
- **openmpi** - the processes are started with the mpirun command
- **intelmpi** - same but for Intel MPI
- **srunmpi** - the processes are started with the srun command

Read more about execution models in the official docs:
https://qcg-pilotjob.readthedocs.io/en/develop/execution_models.html

QCGPilot-Job: Calling for an executable

test.py

```
from qcg.pilotjob.api.manager import LocalManager
from qcg.pilotjob.api.job import Jobs

manager = LocalManager()
jobs = Jobs()

jobs.add(script='echo "job ${it} executed at `date` @ `hostname`"',
        name='test_job',
        stdout='job.out.${it}',
        iteration=4)

job_names = manager.submit(jobs)
print('submited job names: ', str(job_names))
job_status = manager.status(job_names)
print('job status: ', job_status)

manager.wait4all()
manager.finish()
```

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Hands-on

Hands-on #3

Task

- Execute **512 independent short jobs** on two nodes on the “rome” partition. Each node in that partition has 128 cores.

Steps

- Go to **hands-on/example_03**
- Take a look at **qcg_job.py** and **qcg_job.sh**
- Submit the SLURM job script, check the results

Questions

- What host names are reported?
- List the additional files and directories that appear.

```python
from qcg.pilotjob.api.job import Jobs
from qcg.pilotjob.api.manager import LocalManager

# create the QCG manager and the Jobs object
manager = LocalManager()
jobs = Jobs()

# create 512 independent jobs
for job in range(512):
    # define a job
    jobs.add(name="job_{}".format(job),
             exec='hostname',
             stdout='output/job.{}.out'.format(job),
             stderr='output/job.{}.err'.format(job),
             model='default',
             numCores={ "exact" : 1},
             iteration=1)

print("-- submit jobs --")
manager.submit(jobs)
print("-- wait for all jobs --")
manager.wait4all()
manager.finish()
print("-- finished")
```

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Hands-on #4

Task

- Do the task of Hands-on #2 with QCG-PilotJob.
- Aggregate the results in one CSV file.

Steps

- Go to **hands-on/example_04**
- Take a look at **README.md**, **qcg_job_v0.py** and **qcg_job.sh**
- Extract the input data (**input.tar.gz**)
- Submit the SLURM job script, check the results

Steps

- The ***v0** script has a problem. Can you spot it?
- Execute **qcg_job_v1.py**, compare results with the results from **qcg_job_v0.py**

```
job_names = []

# scan input directory for CVS files
for filename in glob(os.path.join("input/*.csv"), recursive = False):
    # determine the job name and append it to the list
    job_names.append(os.path.basename(filename))

# create the QCG manager and the Jobs object
manager = LocalManager()
jobs = Jobs()

# loop over all files and populate the list of jobs
for jname in job_names:
    print("submit {}".format(jname))
    jobs.add(name=jname,
             exec='python3',
             args=["average.py",
            os.path.join("input", jname)],
            stdout='average_{}'.format(jname),
            stderr='job.{}.err'.format(jname),
            modules=[
            "2024", "Python/3.12.3-GCCcore-13.3.0"],
             iteration=1
             )

# add a job for the aggregation of the results
jobs.add(name="aggregate",
         script='cat average_*.csv | sort',
         stdout='result.csv',
         stderr='aggregate.err')
```

Abbreviations

- Message Passing Interface (MPI)
- Shared memory (SHM)
- System billing unit (SBU)



Have fun! :)